

**RESEARCH AND DEVELOPMENT AGREEMENT**

by and between

**HITACHI, LTD.**

and

**OPNEXT, INC.**

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Effective as of October 1, 2001

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## **RESEARCH AND DEVELOPMENT AGREEMENT**

THIS RESEARCH AND DEVELOPMENT AGREEMENT is dated as of July \_\_, 2002, by and between HITACHI, LTD., a corporation existing under the laws of Japan ("Hitachi") and OPNEXT, INC., a Delaware corporation and a Subsidiary of Hitachi ("OpNext"). This Agreement is deemed to be effective as of October 1, 2001 ("Effective Date").

### **RECITALS**

WHEREAS, Hitachi and OpNext Japan, Inc., a corporation existing under the laws of Japan and a Wholly-Owned Subsidiary of OpNext ("OpNext Japan") have entered into a Research and Development Agreement dated July 31, 2001 (as amended, supplemented or otherwise modified from time to time, the "OpNext Japan Agreement"), which provides, *inter alia*, for Hitachi to provide OpNext Japan with certain research and development support; and

WHEREAS, Hitachi and OpNext desire to enter into an agreement, on substantially similar terms and conditions, as applicable, for Hitachi to provide OpNext with certain research and development support.

NOW, THEREFORE, in consideration of the mutual covenants contained herein and other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the parties to this R&D Agreement hereby agree as follows:

**Section 1. Definitions.** The following terms, when used herein with initial capital letters, shall have the respective meanings set forth in this Section 1.

(a) "Affiliate" of any particular Person shall mean any other Person controlling, controlled by or under common control with such particular Person, where "control" means the possession, directly or indirectly, of the power to direct the management and policies of a Person whether through the ownership of voting securities, contract or otherwise.

(b) "Applicable Products" shall have the meaning set forth in Section 12(i) of this R&D Agreement.

(c) "Average Man-Month Cost" shall have the meaning set forth in Exhibit A.

(d) "Base Agreement" shall mean this Research and Development Agreement, excluding any Exhibits and Statements of Work.

(e) "Business" shall mean Hitachi's fiber optic component business of designing, developing, manufacturing, marketing, distributing and selling Products operated by Hitachi's Telecommunications Systems Division as of January 31, 2001 and as operated by OpNext Japan between January 31, 2001 and July 31, 2001.

(f) "Business Day" shall mean any day of the year except Saturday, Sunday and any day on which commercial banking institutions are authorized or obligated by law, regulation or executive order to close in New York or Tokyo.

(g) “Clarity Parties” shall mean, collectively, Clarity Partners L.P., a Delaware limited partnership, Clarity OpNext Holdings I, LLC, a Delaware limited liability company, and Clarity OpNext Holdings II, LLC, a Delaware limited liability company.

(h) “Commercially Reasonable Efforts” shall mean diligent and commercially reasonable and expeditious efforts to accomplish a task or objective in a manner that is at least equal to the efforts, quality and resources devoted by a party that such party would apply to its own high priority task or objective under similar circumstances.

(i) “Confidential Information” shall mean any information not generally known to the public that is (a) made or disclosed in contemplation of this R&D Agreement or (b) information related to the Business that is disclosed or made available to the receiving party pursuant to this R&D Agreement in each case that is stamped or marked as confidential or, in the case of orally conveyed information, is identified at the time of conveyance as confidential and is followed by confirmation in writing reasonably thereafter, including all of the following: (i) prototypes, files, analyses, techniques, systems, formulae, research, records, documentation, models, data, databases, ideas, inventions, designs, developments, devices, methods and processes (whether or not patentable and whether or not reduced to practice); (ii) know-how; (iii) Licensed IP; (iv) Hitachi R&D IP; (v) OpNext R&D IP; and (vi) other Intellectual Property rights. In addition, Confidential Information shall include the terms and conditions of this R&D Agreement.

(j) “Cure Period” shall have the meaning set forth in Section 5(d) of this R&D Agreement.

(k) “Dispute Notice” shall have the meaning set forth in Section 21 of this R&D Agreement.

(l) “Effective Date” shall have the meaning set forth in the preamble of this R&D Agreement.

(m) “Exhibit” shall mean an attachment to this R&D Agreement, as such attachment may be amended from time to time, each one of which is incorporated herein by this reference.

(n) “Existing R&D Agreements” shall have the meaning set forth in Section 3(a) of this R&D Agreement.

(o) “Hitachi R&D IP” shall have the meaning set forth in Section 4(b) of this R&D Agreement.

(p) “Intellectual Property” shall mean all: (i) patents, patent applications, patent disclosures and inventions (including all extensions, reexaminations, reissues, continuations and renewals related thereto); (ii) copyrights (registered or unregistered and all renewals thereof) and copyrightable works and registrations and applications for registration thereof; (iii) mask works and registrations and applications for registration thereof; (iv) computer software, data, databases and documentation thereof; and (v) trade secrets and other confidential information (including ideas, formulas, compositions, inventions (whether patentable or



unpatentable and whether or not reduced to practice), know-how, manufacturing and production processes and techniques, research and development information, drawings, specifications, designs, plans, proposals, technical data, operating, maintenance and safety materials and drawings, test procedures, test data, sources of materials and supplies, financial and marketing plans and customer and supplier lists and information). Intellectual Property, as referred to in this R&D Agreement, refers to rights throughout the world, including any equivalent of any of the foregoing in any jurisdiction or under any laws, regulations or treaties.

(q) "Inventor" shall have the meaning set forth in Section 7 of this R&D Agreement.

(r) "IP License Agreement" shall mean the intellectual property license agreement between Hitachi and OpNext Japan dated July 31, 2001 (as amended, supplemented or otherwise modified from time to time).

(s) "Jointly Developed Intellectual Property" shall mean all Intellectual Property resulting from an R&D Project under this R&D Agreement in accordance with Section 4(c)(ii) hereof, and shall exclude Hitachi R&D IP, OpNext R&D IP and Licensed IP.

(t) "Licensed Hitachi R&D IP" shall have the meaning set forth in Section 5(b) of this R&D Agreement.

(u) "Licensed IP" shall have the meaning set forth in Section 3(a) of the IP License Agreement.

(v) "Losses" shall have the meaning set forth in Section 8(c) of this R&D Agreement.

(w) "Mark-Up" shall have the meaning set forth in Exhibit A.

(x) "Mark-Up Fee" shall mean the fee that Hitachi will charge to OpNext for Hitachi's past investment in Hitachi R&D IP and Jointly Developed Intellectual Property that Hitachi agrees to transfer to OpNext as described in Section 4(c)(iii) and shall be determined in accordance with the formula set forth in Exhibit A.

(y) "Minority-Owned Affiliate" shall mean any entity, that a party, directly or indirectly, at any time, owns or controls twenty percent (20%) to fifty percent (50%) of the voting equity shares or securities convertible into such shares.

(z) "Monthly Cost" shall have the meaning set forth in Exhibit A.

(aa) "New Development Costs" shall mean all of the costs related to a particular R&D Project incurred after commencement of such R&D Project, including operating expenses and charges for the use of any tangible property made available for use in the R&D Project but shall not include the consideration for the use of any existing or underlying Intellectual Property owned or controlled by either party that is used for such R&D Project.

(bb) "Old Development Costs" shall mean all of the costs incurred prior to commencement of a particular R&D Project for development of any existing or underlying Intellectual Property owned or controlled by either party that is used for such R&D Project, including operating expenses and charges for the use of any tangible property made available for use in developing such existing or underlying Intellectual Property.

(cc) "OpNext Japan" shall have the meaning set forth in the first recital of this R&D Agreement.

(dd) "OpNext Japan Agreement" shall have the meaning set forth in the first recital of this R&D Agreement.

(ee) "OpNext R&D IP" shall mean: (i) Intellectual Property that can be clearly identified as that resulting from R&D Projects (excluding any Hitachi R&D IP and Licensed IP) for which OpNext has paid one-hundred percent (100%) of the New Development Costs; and (ii) Jointly Developed Intellectual Property under Section 4(c)(ii) or Hitachi owned Intellectual Property under Section 4(c)(i) that Hitachi has agreed to transfer to OpNext and OpNext has paid a Mark-Up Fee to Hitachi in accordance with Section 4(c)(iii) and Exhibit A.

(ff) "Permitted Entities" shall have the meaning set forth in Section 5(a) of this R&D Agreement.

(gg) "Person" shall mean any individual, corporation, partnership, limited liability company, business trust, association, joint stock company, trust, unincorporated organization, joint venture, firm or other entity or a government or any political subdivision or agency, department or instrumentality thereof.

(hh) "Products" shall mean, collectively, transmitters, receivers, transceivers, laser diode modules, photo diode modules, parallel optical interconnectors, lasers, photodiodes, modulators, amplifier modules, optical switches and optical wave guides.

(ii) "Project Manager" shall have the meaning set forth in Exhibit B.

(jj) "R&D Agreement" shall mean, collectively, the Base Agreement, including any Exhibits and Statements of Work (as amended, supplemented or otherwise modified from time to time).

(kk) "R&D Plan" shall mean the plan prepared jointly by Hitachi and OpNext and which, at a minimum, includes the information set forth in Section 4 of Exhibit B.

(ll) "R&D Procedures" shall mean the procedures set forth in Exhibit B, which may be amended from time to time based upon the parties' mutual agreement.

(mm) "R&D Projects" shall mean research and development projects undertaken by OpNext and/or its Affiliates (other than OpNext Japan) and/or by Hitachi on behalf of OpNext after the Effective Date.

(nn) "R&D Support" shall mean research and development support provided by Hitachi to OpNext.

(oo) "Related Parties" shall have the meaning set forth in Section 8(c).

(pp) "Statement of Work" shall mean a statement of work in substantially the form set forth in Exhibit E for each R&D Project for which Hitachi will provide R&D Support setting forth additional terms and conditions relating to such R&D Project.

(qq) "Subsidiary" shall mean, with respect to any Person, any corporation, limited liability company, partnership, association or other business entity of which (i) if a corporation, a majority of the total voting power of shares of stock entitled (without regard to the occurrence of any contingency) to vote in the election of directors, managers or trustees thereof is at the time owned or controlled, directly or indirectly, by that Person or one or more of the other Subsidiaries of that Person or a combination thereof, or (ii) if a limited liability company, partnership, association or other business entity, a majority of the limited liability company, partnership or other similar ownership interest thereof is at the time owned or controlled, directly or indirectly, by any Person or one or more Subsidiaries of that Person or a combination thereof. For purposes hereof, a Person or Persons shall be deemed to have a majority ownership interest in a limited liability company, partnership, association or other business entity if such Person or Persons shall be allocated a majority of limited liability company, partnership, association or other business entity if such Person or Persons shall be allocated a majority of limited liability company, partnership, association or other business entity gains or losses or shall be or control the managing director or general partner of such limited liability company, partnership, association or other business entity.

(rr) "Total Project Cost" shall have the meaning set forth Exhibit A.

(ss) "Wholly-Owned Subsidiary" shall mean, with respect to any Person, any corporation, limited liability company, partnership, association or other business entity of which (i) if a corporation, one-hundred percent (100%) of the total voting power of shares of stock entitled (without regard to the occurrence of any contingency) to vote in the election of directors, managers or trustees thereof is at the time owned or controlled, directly or indirectly, by that Person or one or more of the other Wholly-Owned Subsidiaries of that Person or a combination thereof, or (ii) if a limited liability company, partnership, association or other business entity, all of the limited liability company, partnership or total ownership interest thereof is at the time owned or controlled, directly or indirectly, by any Person or one or more other Wholly-Owned Subsidiaries of that Person or a combination thereof.

**Section 2. Research and Development Services.** During the term of this R&D Agreement, OpNext may request Hitachi to provide R&D Support for an R&D Project. To the extent such R&D Project is directly related to the Business, Hitachi will provide R&D Support, as requested, to OpNext and the parties will execute an applicable Statement of Work for such R&D Project. If OpNext requests R&D Support for an R&D Project that is not directly related to the Business, Hitachi agrees to negotiate with OpNext in good faith regarding such request; provided, however, if the parties cannot agree as to such R&D Project or an applicable Statement of Work, Hitachi shall have no further obligation with respect to such specific request. Each Statement of

Work shall be separately executed by the parties, and shall be attached to this R&D Agreement as a Schedule to Exhibit E. Hitachi will provide R&D Support for each R&D Project set forth in a Statement of Work in accordance with the R&D Procedures and the other terms and conditions set forth herein. Hitachi will provide such R&D Support consistent with the following:

(a) Meetings. Hitachi and OpNext will hold quarterly joint review meetings to determine and update the R&D Plan for R&D Projects.

(b) Request and Forecasts. Based on the R&D Plan developed in accordance with the R&D Procedures, OpNext will require assistance on a binding basis from Hitachi on specific R&D Projects for the upcoming twelve (12) month period. OpNext will also provide Hitachi with a non-binding forecast of R&D Projects that OpNext expects to request from Hitachi for the twelve (12) month period following such twelve (12) month period. Hitachi and OpNext will update such forecasts during the quarterly meetings described in Section 2(a) above.

(c) Support. Hitachi will be obligated to use Commercially Reasonable Efforts to provide continuous R&D Support for the R&D Projects in accordance with the specific binding requests (to be performed in the upcoming twelve (12) month period) from OpNext and for all specific non-binding projects that OpNext forecasts to be requested from Hitachi on a binding basis within the twelve (12) month period following the end of such twelve (12) month period. Hitachi will use Commercially Reasonable Efforts to be able to support the balance of the non-binding forecast.

### **Section 3. Exclusions from Research and Development Services.**

(a) Hitachi. Nothing contained in this R&D Agreement shall limit in any way Hitachi's ability to continue to conduct research and development activities for other Hitachi business units, including its Affiliates and Subsidiaries, including any fiber optical component business (e.g., semiconductors and cable) subject to the Nonsolicitation or Noncompetition provision in Section 12 of the Stockholders' Agreement between OpNext and each of Hitachi and the Clarity Parties dated July 31, 2001 (as amended, supplemented or otherwise modified from time to time); provided, however, the terms and conditions of this R&D Agreement shall be subject to the terms and conditions of any existing agreements related to the governmental R&D projects, the joint R&D projects with national and public universities or private universities, the R&D projects requested by other Hitachi Subsidiaries or the joint R&D projects with any other agency or organization (collectively, the "Existing R&D Agreements"). Prior to the commencement of any R&D Project, Hitachi shall disclose to OpNext any restrictions contained in the Existing R&D Agreements related to such R&D Project. Hitachi shall use Commercially Reasonable Efforts to obtain consents from applicable parties to the Existing R&D Agreements and from applicable parties to any future R&D agreements so as to avoid limiting or restricting the R&D Projects for which Hitachi agrees to provide R&D Support pursuant to a Statement of Work.

(b) OpNext. Nothing in this R&D Agreement shall in any way limit OpNext's ability to conduct its own, or utilize other third parties to conduct on its behalf, research and development projects.

#### **Section 4. Ownership of intellectual Property Right.**

(a) OpNext's R&D Intellectual Property. OpNext will solely own all right, title and interest, throughout the world in and to OpNext R&D IP. OpNext shall have the right to apply, in its own name and at its own expense, for Intellectual Property protection in the OpNext R&D IP and, if requested, Hitachi shall cooperate with OpNext in any reasonable manner in obtaining such protection, including obtaining signatures of Hitachi Inventors and/or officials on official papers.

(b) Hitachi's R&D Intellectual Property. Hitachi will solely own all Intellectual Property rights that result from all of its other research and development projects, including the Intellectual Property referenced in Section 4(c)(i), excluding Jointly Developed Intellectual Property and/or OpNext R&D IP ("Hitachi R&D IP"). Hitachi shall have the right to apply, in its own name and at its own expense, for Intellectual Property protection in the Hitachi R&D IP and, if requested, OpNext shall cooperate with Hitachi in any reasonable manner in obtaining such protection, including obtaining signatures of OpNext Inventors and/or officials on official papers.

(c) Jointly Developed Intellectual Property. All right, title and interest in and to Jointly Developed Intellectual Property, other than OpNext R&D IP and Hitachi R&D IP, shall be determined in accordance with this Section 4(c).

(i) Hitachi Owned. If the R&D Project is jointly funded by OpNext and Hitachi, unless (ii) or (iii) below applies, the resulting Intellectual Property will be treated as Hitachi R&D IP in accordance with Section 4(b) and will be solely owned by Hitachi.

(ii) Jointly Owned.

(1) If the R&D Project is jointly funded by the parties and either: (a) OpNext contributes fifty percent (50%) or more of the New Development Costs to the R&D Project; or (b) OpNext contributes less than fifty percent (50%) of the New Development Costs to the R&D Project but the parties determine through good faith negotiations that OpNext contributed to the R&D Project in some other fashion, and in both (a) and (b) above the resulting Intellectual Property can clearly be identified with reasonable certainty as that resulting from such R&D Project, then such Intellectual Property shall be deemed Jointly Developed Intellectual Property and shall be owned jointly by the parties and either party may practice such Jointly Developed Intellectual Property without an accounting or compensation to, or the consent of, the other party. Except as set forth in Section 4(c)(iii) below, if either party desires to license any of its rights to the Jointly Developed Intellectual Property herein to a third party, it shall obtain the prior written consent of the other party hereto. Each party shall have the right to apply, in both parties' names, for Intellectual Property protection in the Jointly Developed Intellectual Property, subject to the following: (i) the parties shall agree on the proper way and strategy for proceeding with all protection of the Jointly Developed Intellectual Property in accordance with the R&D Procedures; (ii) all expenses incurred in obtaining and maintaining Intellectual Property protection in the Jointly Developed Intellectual Property shall be equally shared by the parties; and (iii) in the event that one (1) of the parties elects not to seek or maintain patent or other Intellectual Property protection for any Jointly Developed Intellectual

Property in any particular country or not to share equally in the expenses thereof with the other party, the other party shall have the right to seek or maintain such protection at its sole expense in such country and shall have full control over the prosecution and maintenance thereof even though title to any patent or other Intellectual Property protection issuing therefrom shall be jointly owned by the parties.

(2) To the extent OpNext shares the costs of its proportion of the joint funding as described in (1) above, with any Wholly-Owned Subsidiary of OpNext for an R&D Project, OpNext shall have the right to license any Jointly Developed Intellectual Property arising from such R&D Project and the right to sublicense any Hitachi R&D IP associated with such Jointly Developed Intellectual Property, to such Wholly-Owned Subsidiary of OpNext; provided, however, the following conditions are met: (i) OpNext obtains Hitachi's reasonable prior consent; and (ii) such Wholly-Owned Subsidiary of OpNext abides by the terms and conditions of this R&D Agreement. Notwithstanding the foregoing, if any such license invokes any Japanese tax issues then Hitachi shall not be obliged to consent to such license to such Wholly-Owned Subsidiary of OpNext without entering into a separate agreement with such Wholly-Owned Subsidiary of OpNext under reasonable terms and conditions to be agreed upon between the relevant parties to address such tax issues. Notwithstanding the foregoing, such Wholly-Owned Subsidiary of OpNext shall not have any ownership rights in such Jointly Developed Intellectual Property.

(iii) OpNext Owned.

(1) If OpNext desires and Hitachi agrees in its reasonable discretion, OpNext may purchase the Intellectual Property resulting from an R&D Project that is either owned by Hitachi under Section 4(c)(i) or jointly owned by the parties under Section 4(c)(ii), but excluding other Hitachi R&D IP and Licensed IP, by reimbursing Hitachi for any New Development Costs incurred by Hitachi in such R&D Project and paying a Mark-Up to Hitachi in accordance with the formula set forth in Exhibit A hereto.

(2) If OpNext desires and Hitachi agrees in its reasonable discretion, OpNext may purchase the Intellectual Property resulting from an R&D Project that is either owned by Hitachi under Section 4(c)(i) or jointly owned by the parties under Section 4(c)(ii), including the other Hitachi R&D IP and Licensed IP on which such Intellectual Property is based or derived, by paying a Mark-Up Fee. For the purposes of determining such Mark-Up Fee, the parties shall consider the extent of Hitachi's New Development Costs, Hitachi's Old Development Costs and the fair market value of such technology (other than Licensed IP).

(d) Ownership Determination. Prior to the commencement of an R&D Project, the Hitachi and OpNext Project Managers shall discuss in good faith the ownership of the Intellectual Property resulting from such R&D Project based upon the principles listed above. If the parties' Project Managers cannot agree on the ownership of the Intellectual Property, the management of both parties shall discuss in good faith the ownership of the Intellectual Property resulting from such R&D Project. If the management is unable to come to an agreement on such ownership issues (including clear identification of the Intellectual Property resulting from such R&D Project, New Development Costs, Old Development Costs, OpNext's non-monetary

contribution to the R&D Project and the Mark-Up Fee, if applicable), the parties shall refer solely this issue to arbitration pursuant to the arbitration procedures set forth in Exhibit C hereto. In the event that it is impractical to resolve the disputed issue prior to the commencement of the R&D Project (e.g., the parties are unsure as to what extent underlying technology will be utilized or cannot determine the identification of Intellectual Property resulting from the R&D Project, the New Development Costs, Old Development Costs, OpNext's non-monetary contribution to the R&D Project or Mark-Up Fee), either party may elect to proceed with the R&D Project and defer resolution of the disputed issue until a later date; provided that if the parties remain in disagreement after such later date, the parties shall then refer the issue to arbitration as described above. In the event that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C.

**Section 5. Cross License of Intellectual Property.** The parties acknowledge and agree that the license terms set forth in Sections 5(a) and 5(b) below shall apply to OpNext R&D IP and Licensed Hitachi R&D IP relevant to an R&D Project unless the parties set forth different license terms in the applicable Statement of Work. For those R&D Projects that are directly related to the Business, the relevant OpNext R&D IP and Licensed Hitachi R&D IP shall both be licensed on terms no less favorable than the terms set forth in Sections 5(a) and 5(b), respectively, and such licenses shall be set forth in the applicable Statement of Work. For those R&D Projects that are not directly related to the Business for which the parties execute a Statement of Work, the relevant OpNext R&D IP and Licensed Hitachi R&D IP shall both be licensed on terms and conditions to be negotiated by the parties on a case-by-case basis, and such licenses shall be set forth in the applicable Statement of Work.

(a) OpNext R&D IP License. OpNext will license, and does hereby license, the OpNext R&D IP to Hitachi and its Wholly-Owned Subsidiaries on a fully paid-up, non-exclusive, perpetual and irrevocable basis, to use, make, have made, sell, advertise, offer to sell, lease, import, export and supply products and services throughout the world, provided, however, that Hitachi and its Wholly-Owned Subsidiaries shall not have the right to sublicense any OpNext R&D IP to any entity without the consent of OpNext, such consent not to be unreasonably withheld, delayed or conditioned. In addition, OpNext will grant and does hereby grant a right and license to the OpNext R&D IP solely related to the R&D Projects set forth on Exhibit D to solely those entities listed on Exhibit D as licensees to the OpNext R&D IP for such R&D Project (the "Permitted Entities") on a fully paid-up, non-exclusive, perpetual and irrevocable basis, to use, make, have made, sell, advertise, offer to sell, lease, import, export and supply products and services throughout the world; provided, however, that the Permitted Entities shall not have the right to sublicense any such OpNext R&D IP to any entity without the consent of OpNext, such consent not to be unreasonably withheld, delayed or conditioned. Subject to OpNext's consent, which shall not be unreasonably withheld, delayed or conditioned, for each new R&D Project, the parties shall amend Exhibit D to include such new R&D Project and the entities corresponding to such R&D Project that shall be licensed pursuant to this Section 5(a).

(b) Hitachi R&D IP License. Hitachi will license, and does hereby license, the Hitachi R&D IP relevant to (i) R&D Projects directly related to the Business and (ii) R&D Projects that are not directly related to the Business but for which Hitachi agrees to provide R&D Support pursuant to a Statement of Work to OpNext and its Wholly-Owned Subsidiaries on a

fully paid-up, non-exclusive, perpetual and irrevocable basis, to use, make, have made, sell, advertise, offer to sell, lease, import, export and supply products and services throughout the world ("Licensed Hitachi R&D IP"); provided, however, that OpNext and its Wholly-Owned Subsidiaries shall not have the right to sublicense any Licensed Hitachi R&D IP to any entity, without the consent of Hitachi, such consent not to be unreasonably delayed, withheld or conditioned.

(c) Transfer of Hitachi R&D IP. In the event a Subsidiary or division of Hitachi that is the owner of the Licensed Hitachi R&D IP is sold or otherwise transferred by Hitachi, Hitachi will make necessary arrangements to secure a license under the terms and conditions set forth in the applicable Statement of Work for OpNext from the new owner such that OpNext and its Wholly-Owned Subsidiaries can continue to use such Licensed Hitachi R&D IP on terms comparable to those therein until such Licensed Hitachi R&D IP expires.

(d) Termination Conditions. Such license of OpNext R&D IP to Hitachi and of Licensed Hitachi R&D IP to OpNext may be terminated upon notice by the licensor only if: (i) the licensee has committed a material breach of its obligations under this R&D Agreement, the licensor has given written notice of such breach to the licensee and such breach remains uncured after sixty (60) days of receiving notice of such breach (the "Cure Period"), or, in the case of a breach which cannot be cured within such Cure Period, the licensee has not instituted within such Cure Period steps necessary to remedy the default and/or thereafter has not diligently pursued the same to completion, within the Cure Period; or (ii) the breaching party has committed an incurable material breach. In the event the breach is a curable breach that cannot be cured within the Cure Period but the licensee has instituted steps necessary to remedy the default and is thereafter diligently pursuing such cure, both parties shall negotiate to determine whether further pursuit of the cure is reasonable. If the parties cannot agree on a resolution in such negotiations, then this issue shall be referred to arbitration pursuant to the arbitration procedures set forth in Exhibit C hereto to decide whether such breach can be cured or any other alternative remedy should be adopted. In the event the breach is an incurable breach, the parties agree that the matter shall be referred to arbitration pursuant to the arbitration procedures set forth in Exhibit C hereto to determine the appropriate remedy. In the event that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C.

(e) Range of License. The obligations of a party in this Section 5 to license Intellectual Property to the other party shall not apply to any Intellectual Property created or acquired by either party after the earlier of (i) July 31, 2011 or (ii) if this R&D Agreement is terminated pursuant to an arbitration in accordance with Section 21 or upon the agreement of the parties, the effective date of such termination; provided, however, that the licenses under OpNext R&D IP and Licensed Hitachi R&D IP existing as of the earlier of (i) July 31, 2011 or (ii) the effective date of such termination shall continue, under reasonable terms and conditions to be agreed between the parties, until the expiration of all of such OpNext R&D IP and Licensed Hitachi R&D IP.



## **Section 6. Covenants to Protect Intellectual Property.**

(a) Notice of Infringement. If either party learns of facts that may constitute an infringement of any of the Intellectual Property covered by this R&D Agreement or of any allegations that there has been an infringement of such Intellectual Property, it shall promptly notify the owner of the Intellectual Property of such possible infringement. With respect to Jointly Developed Intellectual Property, each party shall notify the other party of such possible infringement. No party shall have any duty to conduct any investigation or to make any inquiry with respect to any such alleged infringement.

(b) Infringement Suits on Jointly Developed Intellectual Property. If both parties agree to initiate appropriate action to cause any infringement of Jointly Developed Intellectual Property to cease, including if necessary bringing suit to enjoin such infringement, the parties shall share the expense and split any damages or other court compensation equally, or in some other proportion to be agreed by the parties prior to initiating an action. If upon notice of infringement of Jointly Developed Intellectual Property, a party elects not to initiate any action, the other party shall have the option to initiate appropriate action to cause any infringement to cease, including if necessary bringing suit to enjoin such infringement; and such party shall then be solely responsible for expenses and shall retain any damages, other court compensation awarded and settlement proceeds, but such party shall not enter into a settlement agreement without the prior written consent of the other party, which shall not be unreasonably withheld, unreasonably delayed or unreasonably conditioned.

(c) Infringement of Licensed Hitachi R&D IP. To the extent a competitor of the Business is infringing the Licensed Hitachi R&D IP in OpNext's reasonable business judgment and such infringement is material to the Business, Hitachi will protect OpNext's interest either by prosecuting the Intellectual Property rights on behalf of OpNext or by taking some other appropriate action that will not have a Material Adverse Effect on the ongoing business of OpNext, provided, however, that any such action taken by Hitachi shall not materially adversely affect any other Affiliates of or business units of Hitachi. Both parties shall consult and cooperate with each other in determining how to respond to the infringing activities. Upon the resolution of such infringement by settlement or otherwise, any damages, profits and awards of whatever nature recoverable for such infringement shall, after deducting the parties' expenses, be reasonably allocated between the parties based on the facts and circumstances of the infringement. Both parties will reasonably consider the option of settling any such matter by granting a sublicense of all or portion of the Licensed Hitachi R&D IP.

**Section 7. Inventor Compensation.** Both parties acknowledge and agree that in the event that any employee of Hitachi or employee of OpNext (hereinafter referred to as the "Inventor") creates any Intellectual Property under this R&D Agreement, the owner of such Intellectual Property shall pay a certain amount of compensation to such Inventor, taking into account the Inventor's contribution and according to the terms and conditions mutually agreed to by the parties.

## **Section 8. Warranties and Limitations.**

(a) Existing R&D Agreements. Hitachi represents and warrants that the terms and conditions of the Existing R&D Agreements shall not have a material impact on OpNext's ability to conduct its research and development activities pursuant to this R&D Agreement or the ownership of or other rights in any Intellectual Property that may result from such activities, and that Hitachi will make Commercially Reasonable Efforts to consult and cooperate with OpNext to eliminate or minimize any negative impact arising from the terms and conditions of any Existing R&D Agreements.

(b) Disclaimer of Warranties. Hitachi expressly disclaims all representations and warranties, express or implied, in connection with the R&D Support provided pursuant to this R&D Agreement, including the warranties of non-infringement and title and the implied warranties of merchantability and fitness for a particular purpose. Such R&D Support is provided on an "as is" basis, except as set forth on Exhibit B.

(c) Indemnification by Hitachi. From and after the Effective Date, Hitachi shall indemnify OpNext and its Affiliates and each of their respective officers, directors, members, stockholders, partners, employees and agents (as applicable) ("Related Parties") and hold them harmless from any loss, liability, settlement, judgment, award, cost, damage or expense (including court costs and reasonable attorneys' fees) (the "Losses") suffered or incurred by any such indemnified party to the extent arising from: (i) any breach of any representation or warranty of Hitachi contained in this R&D Agreement; or (ii) any breach of any covenant of Hitachi contained in this R&D Agreement, except to the extent that OpNext, its Affiliates, their agents and/or their independent contractors have tortiously contributed in an intentional or grossly negligent manner to the event in question. Notwithstanding the foregoing, in no event shall Hitachi indemnify OpNext under this indemnity provision for claims under Section 8(c)(i) brought on or after July 31, 2003.

(d) Indemnification by OpNext. From and after the Effective Date, OpNext shall indemnify Hitachi and its Related Parties and hold them harmless from any Losses suffered or incurred by any such indemnified party to the extent arising from: (i) any breach of any representation or warranty of OpNext contained in this R&D Agreement; or (ii) any breach of any covenant of OpNext contained in this R&D Agreement, except to the extent that Hitachi, its Affiliates, their agents and/or their independent contractors have tortiously contributed in an intentional or grossly negligent manner to the event in question. Notwithstanding the foregoing, in no event shall OpNext indemnify Hitachi under this indemnity provision for claims under Section 8(d)(i) brought on or after July 31, 2003.

(e) IP Infringement Indemnification.

(i) By Hitachi. From and after the Effective Date, Hitachi shall indemnify OpNext and its Wholly-Owned Subsidiaries and their respective Related Parties (each an indemnified party) and hold them harmless from any Losses incurred by any such indemnified party to the extent arising from any claim by a third party that the provision of R&D Support to OpNext by Hitachi under this R&D Agreement and/or that any Licensed Hitachi R&D IP (whether incorporated in an OpNext product or otherwise) infringes or misappropriates such

third party's copyright or trade secret, except to the extent that such claim arises out of or relates to Hitachi's use of, or reliance on, specifications or instructions provided to Hitachi by OpNext.

(ii) By OpNext. From and after the Effective Date, OpNext shall indemnify Hitachi and its Wholly-Owned Subsidiaries and their respective Related Parties (each an indemnified party) and hold them harmless from any Losses incurred by any such indemnified party to the extent arising from any claim by a third party that: (a) the use of the specifications or instructions provided to Hitachi by OpNext in connection with the provision of R&D Support under this R&D Agreement infringes or misappropriates such third party's copyright or trade secret; or (b) the development of a product or the manufacture of a product by OpNext infringes or misappropriates any copyright or trade secret of a third party, except to the extent such infringement or misappropriation is within the scope of Hitachi's indemnification obligations hereunder.

(f) Patent Infringement. If either party receives notice (including, without limitation) a cease and desist letter or invitation to take a license) of a claim or allegation of patent infringement for which, due to the other party's participation in an R&D Project pursuant to this Agreement, such other party may be liable, such party shall promptly notify the other party. Both parties shall consult and cooperate with each other in determining how to respond to such claim or allegation of patent infringement.

(g) Limitation of Liability. Notwithstanding anything to the contrary in this R&D Agreement, neither party's cumulative liability to the other for all claims arising out of or in connection with this R&D Agreement will exceed the fees paid to Hitachi by OpNext pursuant to this R&D Agreement. Neither party shall be liable to the other party or any third party for any special, consequential, exemplary or incidental damages (including lost or anticipated revenues or profits relating to the same), arising from any claim relating to this R&D Agreement, whether such claim is based on warranty, contract, tort (including negligence or strict liability) or otherwise, even if an authorized representative of such party is advised of the possibility or likelihood of same.

**Section 9. Expenses**. OpNext shall be charged for R&D Support on the same basis that Hitachi's Wholly-Owned Subsidiaries are allocated research and development charges for their activities, provided that in no event shall such terms be less advantageous from OpNext's perspective, than those terms which could be reasonably expected to be obtained in an arms-length transaction. Notwithstanding the foregoing, OpNext acknowledges and agrees that if it exercises its rights under Section 4(c)(iii), OpNext shall pay a Mark-Up Fee to Hitachi. Whether or not the transactions contemplated hereby are consummated, and except as otherwise specifically provided in this R&D Agreement, all costs and expenses incurred in connection with this R&D Agreement and the transactions contemplated hereby shall be paid by the party incurring such costs or expenses.

**Section 10. Termination**. This R&D Agreement commences on the date hereof and will automatically terminate and be of no further force or effect upon July 31, 2011.

## **Section 11. Confidentiality.**

(a) Confidentiality Obligations. Confidential Information will not be disclosed or made available by the receiving party, directly or indirectly, to any third party, except as expressly provided herein or as shall be agreed to in writing by the disclosing party. Each of the parties agrees to take all reasonable steps to preserve the confidentiality of the other's Confidential Information in accordance with their respective policies for the protection of their own non-public information (which policies shall provide at least reasonable protection) and agrees that it will be made available only to: (i) those employees as shall have a need to see and use for the purpose of fulfilling that party's obligations under this R&D Agreement, provided that any such employee shall be informed of the confidential nature of the Confidential Information and shall be required to observe the confidentiality obligations in respect thereof; or (ii) its auditors, counsel, other professional advisors, sublicensees authorized under the terms of this R&D Agreement (Section 5(a) for OpNext R&D IP and Section 5(b) for Licensed Hitachi R&D IP) or suppliers, if the receiving party, in its sole discretion, determines that it is reasonably necessary for such Person to have access to such information, provided that any such Person agrees to be bound by the provisions of this Section 11(a) to the same extent as the receiving party. The receiving party shall ensure that all Confidential Information received by it is kept separate (together with all information generated by the receiving party therefrom) from all documents and other records of the receiving party, and that it shall not use, reproduce, transfer or store any of the Confidential Information in an extremely accessible place.

(b) Exclusions.

(i) Confidential Information shall not include (A) any information that has become or previously was generally available to the public other than by reason of a breach of Section 11(a) by the receiving party or has become available to the receiving party on a non-confidential basis after the Effective Date, or (B) any information for which disclosure is required (i) in any report, statement or testimony required by any municipal, state or federal regulatory body having or claiming to have jurisdiction over the receiving party, (ii) in response to any summons or subpoena or in connection with any litigation, or (iii) in order to comply with any law, order, regulation or ruling applicable to the receiving party (it being understood that in each case, to the extent practicable, the receiving party shall provide the disclosing party prompt notice to any such event and cooperate in good faith to enable the disclosing party to participate to protect its interest in such confidential information).

(ii) Notwithstanding Section 11(a), to the extent that after the Effective Date, Hitachi desires to disclose Confidential Information that constitutes OpNext R&D IP to Hitachi Minority-Owned Affiliates and/or suppliers, Hitachi shall notify OpNext of such desire and propose the terms and conditions of an appropriate nondisclosure agreement into which OpNext and the corresponding Hitachi Minority-Owned Affiliate or supplier may enter. OpNext agrees that within fifteen (15) Business Days of receipt of such request and proposed nondisclosure agreement, OpNext shall, at its sole discretion, either: (i) enter into the proposed nondisclosure agreement and directly provide the requested confidential information to the Hitachi Minority-Owned Affiliate or supplier; (ii) propose reasonably modified terms and conditions of the nondisclosure agreement under which OpNext will provide the requested confidential information to Hitachi's Minority-Owned Affiliate or supplier; or (iii) commence

discussions with Hitachi to reach a resolution of OpNext's concerns with respect to such disclosure, if OpNext believes such disclosure is not in the best interest of the parties. In the event that OpNext elects to exercise option (ii) or (iii), the parties agree to negotiate in good faith and on reasonable terms to resolve the situation within a reasonable amount of time, which shall not exceed fifteen (15) Business Days of OpNext's provision of such a response. If the parties cannot agree on a resolution in such negotiations, then this issue shall be referred to arbitration pursuant to the arbitration procedures set forth in Exhibit C hereto. In the event that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C.

(c) Injunctive Relief. The parties acknowledge and agree that money damages would be inadequate to remedy any breach of the confidentiality obligations in this Section 11 and that the non-breaching party shall be entitled to obtain equitable remedies with respect to any such breach, including preliminary injunctive relief in accordance with Section 27.

(d) Ownership. All Confidential Information furnished hereunder shall be and remain the exclusive property of the disclosing party, and the receiving party agrees to promptly return to the disclosing party, upon the disclosing party's request, all documents, samples and other material in the possession, custody or control of the receiving party that bear or incorporate any part of the Confidential Information, including all copies made by the receiving party, except as otherwise provided herein.

(e) Press Releases and Announcements. Each party agrees to consult with the other as to the general nature of any news releases or public statements with respect to the transactions contemplated by this R&D Agreement, and use Commercially Reasonable Efforts not to issue any news releases or public statements inconsistent with results of such consultations. Subject to applicable laws or the rules of any applicable securities exchange, each party shall use Commercial Reasonable Efforts to enable the other party to review and comment on all such news releases prior to the release thereof.

## **Section 12. Export Control.**

(i) Each party represents and warrants that it shall not use any products or software (including technology relating thereto) provided by other party under this R&D Agreement and any other products, software and/or technology manufactured or developed by using such products or software (hereinafter called, "Applicable Products") for the purposes of disturbing international peace and security, including (a) the design, development, production, stockpiling or any use of weapons of mass destruction such as nuclear, chemical or biological weapons or missiles, (b) other military activities, or (c) any use reasonably supporting these activities.

(ii) Each party also represents and warrants that it shall not sell, export, dispose of, license, rent, transfer, disclose or otherwise provide, whether directly or indirectly, Applicable Products to any third party for whom the warranting party has knowledge or reason to know that such third party will engage in the activities described in Section 12(i). Each party shall be responsible for the entities to whom it sells, exports, disposes of, licenses, rents, transfers, discloses or otherwise provides Applicable Products.

(iii) Each party shall not directly or indirectly, export, re-export, transship, or otherwise transfer Applicable Products in violation of any applicable export control laws and regulations promulgated and administered by the governments of the countries asserting jurisdiction over the parties or transactions.

**Section 13. Notices.** Any notice provided for in this R&D Agreement shall be in writing and shall be either personally delivered, mailed first class (postage prepaid) or sent by reputable overnight courier service (charges prepaid) to the parties at the address set forth below or at such address or to the attention of such other person as the recipient party has specified by prior written notice to the sending party. Notices shall be deemed to have been given hereunder on the date delivered when delivered personally, seven (7) days after deposit in the U.S. mail or Japanese mail and three (3) days after deposit with a reputable overnight courier service. The addresses for OpNext and Hitachi are:

If to OpNext:

OpNext, Inc.  
One Christopher Way  
Eatontown, New Jersey 07724  
Attention: Chief Executive Officer  
Fax: (732) 544-3561

with a copy which shall not constitute notice to OpNext, to:

Simpson Thacher & Bartlett  
425 Lexington Avenue  
New York, NY 10017-3954  
Attention: I. Scott Gottdiener, Esq.  
Fax: (212) 455-2502

with a copy which shall not constitute notice to OpNext, to:

Clarity Partners, L.P.  
100 North Crescent Drive  
Beverly Hills, CA 90210  
Attention: David Lee  
Fax: (310) 432-5000

If to Hitachi:

Hitachi, Ltd.  
Research & Development Group  
New Marunouchi Bldg.,  
5-1, Marunouchi 1-chome  
Chiyoda-ku, Tokyo, 100-8220 Japan  
Attention: President  
Fax: 81-3-3214-3349

with a copy, which will not constitute notice to Hitachi, to:

Kirkland & Ellis  
200 East Randolph Drive  
Chicago, IL 60601  
Attention: William A. Streff Jr., Esq.  
Fax: (312) 861-2200

with a copy, which will not constitute notice to Hitachi, to:

Hitachi, Ltd.  
6, Kanda-Surugadai 4-chome  
Chiyoda-ku  
Tokyo, 101-8010 Japan  
Attention: Senior Group Executive, Information & Telecommunication Systems Group  
Fax: 81-3-5295-1563

**Section 14. Amendment and Waiver.** No amendment of any provision of this R&D Agreement shall be valid unless the same shall be in writing and signed by OpNext and Hitachi. The failure of any party to enforce any of the provisions of this R&D Agreement shall in no way be construed as a waiver of such provisions and shall not affect the right of such party thereafter to enforce each and every provision of this R&D Agreement in accordance with its terms.

**Section 15. Assignment.** Except as set forth below, this R&D Agreement and any rights and obligations hereunder shall not be assignable or transferable by OpNext or Hitachi (including by operation of law in connection with a merger or sale of stock, or sale of substantially all the assets, of OpNext or Hitachi) without the prior written consent of the other party and any purported assignment without such consent shall be void and without effect.

**Section 16. Counterparts.** This R&D Agreement may be executed in one or more counterparts, each of which shall be an original and all of which taken together shall constitute one and the same agreement.

**Section 17. Delivery by Facsimile.** This R&D Agreement, the agreements referred to herein, and each other agreement or instrument entered into in connection herewith or contemplated hereby, and any amendments hereto or thereto, to the extent signed and delivered by means of a facsimile machine, shall be treated in all manner and respects as an original agreement or instrument and shall be considered to have the same binding legal effect as if it were the original signed version thereof delivered in person. At the request of any party hereto or to any such agreement or instrument, each other party hereto shall reexecute original forms thereof and deliver them to all other parties. No party hereto or to any such agreement or instrument shall raise the use of a facsimile machine to deliver a signature or the fact that any signature or agreement or instrument was transmitted or communicated through the use of a facsimile machine as a defense to the enforceability of a contract and each such party forever waives any such defense.

**Section 18. Exhibits and Schedules.** All Exhibits and Schedules annexed hereto or referred to herein are hereby incorporated in and made a part of this R&D Agreement as if set forth in full herein.

**Section 19. Further Assurances.** During the term of this R&D Agreement and at all times thereafter, each party shall provide to the other party, at its request, reasonable cooperation and assistance (including the execution and delivery of affidavits, declarations, oaths, assignments, samples, exhibits, specimens and any other documentation) as necessary to effect the terms of this R&D Agreement.

**Section 20. Governing Law.** This R&D Agreement shall be governed by and construed in accordance with the laws of New York without giving effect to any choice-of-law or conflict-of-law provision or rule (whether of New York or any other jurisdiction) that would cause the application of the laws of any jurisdiction other than of New York. Regardless of the law applied, because this contract is in English, the terms and conditions of this contract will be interpreted in accordance with the meaning of the words in American colloquial English, notwithstanding any meaning of any word when translated into its Japanese equivalent.

**Section 21. Dispute Resolution.** All disputes between the parties relating to this R&D Agreement shall be resolved by arbitration. In the event of any dispute under this R&D Agreement, as a condition precedent to either party seeking arbitration, in connection therewith, the parties will attempt to resolve such dispute by good faith negotiations (except for actions seeking injunctive relief). Such negotiations shall first involve the individuals designated by the parties as having general responsibility for the R&D Agreement. If such negotiations do not result within thirty (30) days from written notice of either party indicating that a dispute exists (a "Dispute Notice") in a resolution of the dispute, OpNext shall nominate one (1) corporate officer of the rank of vice president or higher and Hitachi shall nominate one (1) corporate officer of the rank of Board Director or higher, which corporate officers shall meet in person and attempt in good faith to negotiate a resolution to the dispute. In the event the corporate executives are unable to resolve the dispute within forty-five (45) days of receipt by either party of a Dispute Notice, a party may refer the matter to arbitration (except in the case of disputes arising under Section 11(c) for which the parties may seek preliminary injunctive relief pending resolution of the issue hereunder). In the event that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C.

**Section 22. Interpretation.** The headings and captions contained in this R&D Agreement, in any Exhibit or Schedule hereto and in the table of contents to this R&D Agreement are for reference purposes only and do not constitute a part of this R&D Agreement. The use of the word "including" herein shall mean "including without limitation."

**Section 23. No Strict Construction.** Notwithstanding the fact that this R&D Agreement has been drafted or prepared by one of the parties, OpNext and Hitachi confirm that both they and their respective counsel have reviewed, negotiated and adopted this R&D Agreement as the joint agreement and understanding of the parties, and the language used in this R&D Agreement shall be deemed to be the language chosen by the parties hereto to express their mutual intent, and no rule of strict construction shall be applied against any Person.



**Section 24. Recordation.** This R&D Agreement effects a transfer and license of rights in certain Intellectual Property and may be recorded in appropriate recordal repositories to evidence such transfer and license of rights.

**Section 25. Relationship of the Parties.** The parties hereto are independent contractors. The rights, obligations and liabilities of the parties shall be several and not joint or collective and nothing contained in this R&D Agreement shall be construed as creating a partnership, joint venture, agency, employment, trust or other association of any kind, each party being individually and independently responsible as set forth in this R&D Agreement.

**Section 26. Severability.** Whenever possible, each provision of this R&D Agreement shall be interpreted in such manner as to be effective and valid under applicable law, but if any provision of this R&D Agreement is held to be invalid, illegal or unenforceable in any respect under any applicable law or rule in any jurisdiction, such invalidity, illegality or unenforceability shall not affect the validity, legality or enforceability of any other provision of this R&D Agreement in such jurisdiction or affect the validity, legality or enforceability of any provision in any other jurisdiction, but this R&D Agreement shall be reformed, construed and enforced in such jurisdiction as if such invalid, illegal or unenforceable provision had never been contained herein.

**Section 27. Submission to Jurisdiction.** With respect to actions for preliminary injunctive relief sought pursuant to Section 11 or for confirmation or enforcement of an arbitration pursuant to Section 21, each party to this R&D Agreement (including any third-party beneficiaries to this R&D Agreement) hereby irrevocably and unconditionally:

(i) submits for itself and its property to the exclusive jurisdiction of the courts of New York situated in New York City;

(ii) consents that any such action or proceeding may be brought in such courts, and waives any objection that it may now or hereafter have to the venue of any such action or proceeding in any such court or that such action or proceeding was brought in an inconvenient court and agrees not to plead or claim the same;

(iii) agrees that service of process in any such action or proceeding may be effected by mailing a copy thereof by registered or certified mail (or any substantially similar form of mail), postage prepaid, to such party at its address set forth herein or at such other address of which the agent shall have been notified pursuant thereto; and

(iv) agrees that nothing contained herein shall affect the right to effect service of process in any other manner permitted by law.

**Section 28. Survival.** To the extent the terms of this R&D Agreement provide for rights, interest, duties, claims, undertakings and obligations subsequent to the termination or expiration of this R&D Agreement, such terms of this R&D Agreement shall survive such termination or expiration, including but not limited to the terms of Sections 1, 4, 5(e) (with respect to Hitachi R&D IP and/or OpNext R&D IP that has not expired), 6, 8, 9 (with respect to accrued but unpaid expenses), 11, 12, 18, 19, 20, 21, 22, 23, 26-32.

**Section 29. Third-Party Beneficiaries.** OpNext and Hitachi acknowledge and agree that this R&D Agreement is intended only for the benefit of themselves and their Subsidiaries, except, for purposes of Section 11(b)(ii), their Minority-Owned Affiliates, the Clarity Parties, OpNext Japan and OpNext's Subsidiaries and Minority-Owned Affiliates.

**Section 30. Entire Agreement.** Except as otherwise expressly set forth herein, this R&D Agreement embodies the complete agreement and understanding among the parties hereto with respect to the subject matter hereof and supersedes and preempts any prior understandings, agreements or representations by or among the parties, written or oral, which may have related to the subject matter hereof in any way. The parties agree and intend that nothing in this R&D Agreement shall in any way limit or modify any of the terms and conditions of any other agreement between Hitachi and OpNext and/or OpNext Japan, including the OpNext Japan Agreement and agreements related thereto.

**Section 31. Bankruptcy.** The parties agree and intend that the licenses to the Hitachi R&D IP and OpNext R&D IP hereunder shall be considered licenses to "intellectual property" as that term is defined in 11 U.S.C. §101, and that in the event of the bankruptcy of a licensor hereunder, the licensee hereunder shall be entitled to exercise all of the rights and remedies available under 11 U.S.C. §365(n) to the fullest extent permitted by applicable law.

**Section 32. Order of Precedence.** The various parts of this R&D Agreement are intended to be complementary; however, any inconsistency, ambiguity or conflict between this Base Agreement, its Exhibits, the R&D Plan and any Statement of Work shall be resolved in the following order of precedence (with (i) having the highest priority): (i) Base Agreement; (ii) Statement of Work; (iii) Exhibits and (iv) the R&D Plan. Notwithstanding the foregoing, the parties acknowledge that they may agree in an applicable Statement of Work upon specific terms and conditions intended to supersede those contained in this Base Agreement or an Exhibit, provided (x) the Statement of Work makes clear that such is the parties' intent and (y) legal counsel representing each party have reviewed and concurred with such terms and conditions.

**SIGNATURE PAGE TO R&D AGREEMENT**

IN WITNESS WHEREOF, the parties have caused this Agreement to be executed by their duly authorized officers as of the date first written above.

**OPNEXT, INC.**

By: /s/ Minoru Maeda  
Minoru Maeda, Ph.D.  
Board Director  
Chief Operating Officer

**HITACHI, LTD.**

By: /s/ Michiharu Nakamura  
Michiharu Nakamura, Ph.D.  
Corporate Officer and President,  
Research and Development Group

## **FIRST AMENDMENT TO OPNEXT, INC. R&D AGREEMENT**

THIS FIRST AMENDMENT TO OPNEXT, INC. R&D AGREEMENT (the "Amendment") is entered into as of October 1, 2002 (the "Amendment Date"), by and between Hitachi, Ltd., a corporation existing under the laws of Japan ("Hitachi") and OpNext, Inc., a Delaware corporation ("OpNext"). All capitalized terms used herein but not defined herein shall have the meanings ascribed to such terms in the R&D Agreement (as defined below).

### **RECITALS**

WHEREAS, Hitachi and OpNext have entered into that certain R&D Agreement dated as of July 31, 2002 (the "R&D Agreement"); and

WHEREAS, Hitachi and OpNext desire to enter into this Amendment to amend the R&D Agreement as set forth herein.

NOW, THEREFORE, in consideration of the mutual covenants contained herein and other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the parties to this Amendment hereby agree as follows:

#### **Section 1. Amendment Date.**

This Amendment shall be effective as of October 1, 2002. R&D Projects requested by OpNext prior to the Amendment Date shall be governed by the R&D Agreement. R&D Projects requested after the Amendment Date shall be governed by the R&D Agreement as amended by this Amendment. This Amendment and any amendments made to the provisions of the R&D Agreement shall have no retroactive effect.

#### **Section 2. Amendment.**

(a) Section 5(a) of the R&D Agreement is hereby amended by inserting "(i)" after the section heading "OpNext R&D IP License." and before the first sentence of such section and adding the following clause (ii):

(ii) Status of Wholly-Owned Subsidiaries.

(1) License to OpNext R&D IP. If at any time a Wholly-Owned Subsidiary of Hitachi ceases to remain a Wholly-Owned Subsidiary of Hitachi, Hitachi shall provide written notice of such change to OpNext in accordance with Section 13 of this R&D Agreement and the license under OpNext R&D IP existing as of the date such Wholly-Owned Subsidiary ceases to remain a Wholly-Owned Subsidiary, shall continue, pursuant to the terms and conditions of this R&D Agreement; provided, however, for any OpNext R&D IP that is developed after a Wholly-Owned Subsidiary ceases to remain a Wholly-Owned Subsidiary, the

parties shall negotiate in good faith and on commercially reasonable terms a new license governing such OpNext R&D IP.

(2) Sublicenses. For the avoidance of doubt, this R&D Agreement does not grant Wholly-Owned Subsidiaries of Hitachi the right to sublicense the OpNext R&D IP and an entity that ceases to remain a Wholly-Owned Subsidiary of Hitachi shall not have the right to sublicense the OpNext R&D IP without the prior written consent of OpNext, not to be unreasonably withheld, unreasonably delayed or unreasonably conditioned; provided, however, that to the extent any sublicenses have been granted with OpNext's prior written consent with respect to the OpNext R&D IP during the time such entity is a Wholly-Owned Subsidiary of Hitachi, such sublicenses shall continue, pursuant to the terms and conditions of this R&D Agreement and such sublicense.

(b) Section 5(b) of the R&D Agreement is hereby amended by inserting "(i)" after the section heading "Hitachi R&D IP License." and before the first sentence of such section and adding the following clause (ii):

(ii) Status of Wholly-Owned Subsidiaries.

(1) License to Licensed Hitachi R&D IP. If at any time a Wholly-Owned Subsidiary of OpNext ceases to remain a Wholly-Owned Subsidiary of OpNext, OpNext shall provide Hitachi with written notice of such change in accordance with Section 13 of the R&D Agreement and the license under Licensed Hitachi R&D IP existing as of the date such Wholly-Owned Subsidiary ceases to remain a Wholly-Owned Subsidiary, shall continue, pursuant to the terms and conditions of this R&D Agreement; provided, however, for any Hitachi R&D IP that is developed after a Wholly-Owned Subsidiary ceases to remain a Wholly-Owned Subsidiary, the parties shall negotiate in good faith and on commercially reasonable terms a new license governing such Hitachi R&D IP.

(2) Sublicenses. For the avoidance of doubt, this R&D Agreement does not grant Wholly-Owned Subsidiaries of OpNext the right to sublicense the Licensed Hitachi R&D IP and an entity that ceases to remain a Wholly-Owned Subsidiary of OpNext shall not have the right to sublicense the Licensed Hitachi R&D IP without the prior written consent of Hitachi, not to be unreasonably withheld, unreasonably delayed or unreasonably conditioned; provided, however, that to the extent any sublicenses have been granted with Hitachi's prior written consent with respect to the Licensed Hitachi R&D IP during the time such entity is a Wholly-Owned Subsidiary of OpNext, such sublicenses shall continue, pursuant to the terms and conditions of this R&D Agreement and such sublicense.

(c) Section 6 of the R&D Agreement is hereby amended by deleting clause (c) it in its entirety and replacing it with the following clause (c):

(c) Infringement of Licensed Hitachi R&D IP. To the extent a competitor of the Business is infringing the Licensed Hitachi R&D IP in OpNext's reasonable business judgment and such infringement is material to the Business, Hitachi, in its sole discretion, will protect OpNext's interest by either: (i) initiating and maintaining legal proceedings with respect to such alleged infringement or misappropriation against any such Person on behalf of OpNext or (ii) by taking some other appropriate action that will not have a Material Adverse Effect on the ongoing business of OpNext; provided that with respect to both (i) and (ii) both parties shall consult and cooperate with each other in determining how to respond to the infringing activities. For the avoidance of doubt, Hitachi may or may not consult with OpNext prior to determining whether to pursue (i) or (ii). Upon the resolution of such infringement by settlement or otherwise, any damages, profits and awards of whatever nature recoverable for such infringement shall, after deducting the parties' expenses, be reasonably allocated between the parties based on the facts and circumstances of the infringement. Both parties will reasonably consider the option of settling any such matter by granting a sublicense of all or portion of the Licensed Hitachi R&D IP.

(c): (d) Section 6 of the R&D Agreement is hereby amended by adding the following clause (d) after clause

(d) Guaranty.

(i) Hitachi.

(1) Hitachi will use reasonable best efforts to cause its Wholly-Owned Subsidiaries (for so long as they are Wholly-Owned Subsidiaries) to comply with the terms and conditions of this R&D Agreement and Hitachi shall be liable for any breach of such terms and conditions.

(2) Hitachi will use reasonable best efforts to cause its Wholly-Owned Subsidiaries that cease to remain Wholly-Owned Subsidiaries to comply with the terms and conditions of this R&D Agreement applicable to such entities and Hitachi shall be liable for any breach of such terms and conditions.

(ii) OpNext.

(1) OpNext will use reasonable best efforts to cause its Wholly-Owned Subsidiaries (for so long as they are Wholly-Owned Subsidiaries) to comply with the terms and conditions of

this R&D Agreement and OpNext shall be liable for any breach of such terms and conditions.

(2) OpNext will use reasonable best efforts to cause its Wholly-Owned Subsidiaries that cease to remain Wholly-Owned Subsidiaries to comply with the terms and conditions of this R&D Agreement applicable to such entities and OpNext shall be liable for any breach of such terms and conditions.

(e) This R&D Agreement is hereby amended by adding the following clause 33:

Section 33. Hitachi Communication Technologies. For purposes of this R&D Agreement, the defined term "Wholly-Owned Subsidiary" shall not include Hitachi's Wholly-Owned Subsidiary, Hitachi Communication Technologies, Ltd.

### **Section 3. No Other Amendments.**

Except as expressly set forth herein, all other terms and conditions of the R&D Agreement shall remain unmodified, in full force and effect and shall apply to this Amendment.

### **Section 4. Governing Law.**

This Amendment shall be governed by and construed in accordance with the laws of New York without giving effect to any choice-of-law or conflict-of-law provision or rule (whether of New York or any other jurisdiction) that would cause the application of the laws of any jurisdiction other than of New York. Regardless of the law applied, because this contract is in English, the terms and conditions of this contract will be interpreted in accordance with the meaning of the words in American colloquial English, notwithstanding any meaning of any word when translated into its Japanese equivalent.

### **Section 5. Dispute Resolution.**

All disputes between the parties relating to this Amendment shall be resolved by arbitration. In the event of any dispute under this Amendment, as a condition precedent to either party seeking arbitration, in connection therewith, the parties will attempt to resolve such dispute by good faith negotiations (except for actions seeking injunctive relief). Such negotiations shall first involve the individuals designated by the parties as having general responsibility for the R&D Agreement. If such negotiations do not result within thirty (30) days from written notice of either party indicating that a dispute exists (a "Dispute Notice") in a resolution of the dispute, Opto-Device shall nominate one (1) corporate officer of the rank of vice president or higher and Hitachi shall nominate one (1) corporate officer of the rank of Board Director or higher, which corporate officers shall meet in person and attempt in good faith to negotiate a resolution to the dispute. In the event the corporate executives are unable to resolve the dispute

within forty-five (45) days of receipt by either party of a Dispute Notice, a party may refer the matter to arbitration (except in the case of disputes arising under Section 11(c) of the R&D Agreement for which the parties may seek preliminary injunctive relief pending resolution of the issue hereunder). In the event that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C to the R&D Agreement.

**Section 6. Interpretation.**

The headings and captions contained in this Amendment and in any Exhibit are for reference purposes only and do not constitute a part of this Amendment. The use of the word “including” herein shall mean “including without limitation.”

**Section 7. Severability.**

Whenever possible, each provision of this Amendment shall be interpreted in such manner as to be effective and valid under applicable law, but if any provision of this Amendment held to be invalid, illegal or unenforceable in any respect under any applicable law or rule in any jurisdiction, such invalidity, illegality or unenforceability shall not affect the validity, legality or enforceability of any other provision of this Amendment in such jurisdiction or affect the validity, legality or enforceability of any provision in any other jurisdiction, but this Amendment shall be reformed, construed and enforced in such jurisdiction as if such invalid, illegal or unenforceable provision had never been contained herein.

**Section 8. Counterparts.**

This Amendment may be executed in one or more counterparts, each of which shall be an original and all of which taken together shall constitute one and the same agreement.

\* \* \* \* \*



**SIGNATURE PAGE TO OPI R&D AMENDMENT**

IN WITNESS WHEREOF, the parties have caused this Amendment to be executed by their duly authorized officers as of the Amendment Date.

**OPNEXT, INC.**

By: /s/ Harry L. Bosco  
Harry L. Bosco  
Chief Executive Officer and President

**HITACHI, LTD.**

By: /s/ Michiharu Nakamura  
Michiharu Nakamura, Ph.D  
Corporate Officer and President,  
Research and Development Group

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## **SECOND AMENDMENT TO OPNEXT, INC. R&D AGREEMENT**

THIS SECOND AMENDMENT TO OPNEXT, INC. R&D AGREEMENT (this "Amendment") is entered into as of October 27, 2006 (the "Amendment Date"), by and between Hitachi, Ltd., a corporation existing under the laws of Japan ("Hitachi") and Opnext, Inc., a Delaware corporation ("Opnext"). All capitalized terms used herein but not defined herein shall have the meaning ascribed to such terms in the R&D Agreement (as defined below).

### **RECITALS**

WHEREAS, Hitachi and Opnext have entered into that certain Research and Development Agreement dated as of October 1, 2001 (the "Original R&D Agreement"), as amended by the First Amendment thereto dated as of October 1, 2002 (the "First Amendment" and together with any other amendments to the Original R&D Agreement, the "R&D Agreement"); and

WHEREAS, Hitachi and Opnext desire to enter into this Amendment to further amend the R&D Agreement as set forth herein.

NOW, THEREFORE, in consideration of the mutual covenants contained herein and other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the parties to this Amendment hereby agree as follows:

#### **Section 1. Amendment Date.**

This Amendment shall be effective as of the Amendment Date. This Amendment and any amendments made to the provisions of the R&D Agreement hereunder shall have no retroactive effect.

#### **Section 2. Amendment.**

(1) The last sentence of Section 4(a) of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following sentence:

Opnext shall have the right to apply, in its own name and at its own expense, for Intellectual Property protection in Opnext R&D IP and, if requested, Hitachi shall cooperate with Opnext in a reasonable manner in obtaining such protection, including, obtaining signatures of Hitachi employees, contractors and/or officials on official papers.

(2) The last sentence of Section 4(b) of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following sentence:

Hitachi shall have the right to apply, in its own name and at its own expense, for Intellectual Property protection in Hitachi R&D IP and, if requested, Opnext shall cooperate with Hitachi in any reasonable manner in

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obtaining such protection, including, obtaining signatures of Opnext employees, contractors and/or officials on official papers.

(3) Section 4(c)(ii)(1) of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following Section 4(c)(ii)(1):

(1) If the R&D Project is jointly funded by the parties and either: (A) Opnext contributes fifty percent (50%) or more of the New Development Costs to the R&D Project; or (B) Opnext contributes less than fifty percent (50%) of the New Development Costs to the R&D Project but (i) one or more employees or contractors (other than employees of Hitachi or its Subsidiaries) of Opnext are “inventors” under the applicable patent laws of the applicable jurisdiction or (ii) the parties determine through good faith negotiations that Opnext contributed to the R&D Project in some other fashion, and in both (A) and (B) above the resulting Intellectual Property can clearly be identified with reasonable certainty as that resulting from such R&D Project, then such Intellectual Property shall be deemed Jointly Developed Intellectual Property and shall be owned jointly by the parties and either party may practice such Jointly Developed Intellectual Property without an accounting or compensation to, or the consent of, the other party. Except as set forth in Section 4(c)(iii) below, if either party desires to license any of its rights to the Jointly Developed Intellectual Property herein to a third party, it shall obtain the prior written consent of the other party hereto. Each party shall have the right to apply, in both parties’ names, for Intellectual Property protection in the Jointly Developed Intellectual Property. The parties shall agree on the proper way and strategy for proceeding with all protection of the Jointly Developed Intellectual Property in accordance with the R&D Procedures. All expenses incurred in obtaining and maintaining Intellectual Property protection in the Jointly Developed Intellectual Property shall be equally shared by the parties. In the event that one (1) of the parties elects not to seek or maintain patent or other intellectual or industrial property protection for any Jointly Developed Intellectual Property in any particular country or not to share equally in the expenses thereof with the other party, the other party shall have the right to seek or maintain such protection at its sole expense in such country and shall have full control over the prosecution and maintenance thereof even though title to any patent or other intellectual or industrial property protection issuing therefrom shall be jointly owned by the parties.

(4) Section 5(d) of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following Section 5(d):

(d) Termination Conditions. Such a license of Opnext R&D IP to Hitachi and of Licensed Hitachi R&D IP to Opnext shall not be terminated or its exploitation enjoined, until and unless: (i) the licensee has committed a material breach of its obligations under this R&D Agreement, the licensor has given written notice of such breach to the licensee and such breach remains uncured after sixty (60) days of receiving notice of such breach (the “Cure

Period”), or, in the case of a breach which cannot be cured within such Cure Period, the licensee has not instituted within such Cure Period steps necessary to remedy the default and/or thereafter has not diligently pursued the same to completion; or (ii) the breaching party has committed an incurable material breach. In the event the breach is a curable breach that cannot be cured within the Cure Period but the licensee has instituted steps necessary to remedy the default and is thereafter diligently pursuing such cure, both parties shall negotiate to determine whether further pursuit of the cure is reasonable. If the parties cannot agree on a resolution in such negotiations, then this issue shall be referred to arbitration pursuant to the arbitration procedures set forth in Exhibit C hereto to decide whether such breach can be cured or any other alternative remedy should be adopted. In the event the breach is an incurable breach, (i) the parties agree that the matter shall be referred to arbitration pursuant to the arbitration procedures set forth in Exhibit C hereto to determine the appropriate remedy, and (ii) the breaching party shall provide an on-going plan to address the prevention of such a breach occurring again reasonably acceptable to non-breaching party within sixty (60) days of written notice of the breach and shall implement and comply with such plan within the time period set forth in such plan. In the event that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C.

(5) Section 5(e) of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following Section 5(e):

(e) License Term and Review of Obligations.

(i) License Term. The license to the Opnext R&D IP to Hitachi and Licensed Hitachi Future R&D IP (as defined below) to Opnext shall be irrevocable and: (i) with respect to patent rights, shall survive for so long as any applicable patent is valid; and (ii) with respect to all other Opnext R&D IP and Licensed Hitachi Future R&D IP, shall be perpetual. For purposes of this Section, the term “Licensed Hitachi Future R&D IP” means Licensed Hitachi R&D IP resulting from a R&D Project. Notwithstanding the foregoing, (A) if one (1) of the conditions set forth in Section 5(d) is met, (x) Hitachi may terminate the licenses to Licensed Hitachi R&D IP that is developed or filed on or after the effective date of termination and (y) the licenses granted Opnext to Licensed Hitachi Future R&D IP developed or filed prior to the effective date of termination shall continue pursuant to the terms and conditions set forth herein and (B) if one (1) of the conditions set forth in Section 5(d) is met, (x) Opnext may terminate the licenses to Opnext R&D IP that is developed or filed on or after the effective date of termination and (y) the licenses granted Hitachi to Opnext R&D IP developed or filed prior to the effective date of termination shall continue pursuant to the terms and conditions set forth herein.

(ii) Review of Obligations. The obligations set forth in this Section 5 with respect to Licensed Hitachi Other R&D IP (as defined below) shall expire on July 31, 2011; provided, however, that the licenses under Licensed

Hitachi Other R&D IP existing as of July 31, 2011 shall continue, under reasonable terms and conditions to be agreed between the parties, until the expiration of all of such Licensed Hitachi Other R&D IP. Notwithstanding the foregoing, if one (1) of the conditions set forth in Section 5(d) is met, Hitachi may elect to be completely relieved of its obligations set forth in this Section 5 with respect to Licensed Hitachi Other R&D IP. If Hitachi elects to be relieved of its obligations under this Section 5(d), the parties shall renegotiate in good faith and on commercially reasonable terms a new license governing the Licensed Hitachi Other R&D IP. For purposes of this Section, the term "Licensed Hitachi Other R&D IP" means any Licensed Hitachi R&D IP other than Licensed Hitachi Future R&D IP. For the avoidance of doubt, the expiration or termination of Opnext's rights under this R&D Agreement with respect to Licensed Hitachi Other R&D IP will in no way affect Opnext's rights with respect to such Licensed Hitachi Other R&D IP, if any, under any other agreement to which Opnext is a party.

(6) Section 10 of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following:

This R&D Agreement commences on the date hereof and will automatically terminate and be of no further force or effect upon the fifth (5th) anniversary of an Initial Public Offering (as defined in the Stockholders' Agreement between Opnext and Hitachi, Clarity Partners, L.P., Clarity Opnext Holdings I, LLC and Clarity Opnext Holdings II, LLC dated July 31, 2001 (together with any amendments thereto, the "Stockholders' Agreement")); provided, however, that the provisions of this R&D Agreement identified in Section 28 shall survive expiration or termination of this R&D Agreement.

(7) Section 11(a) of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following Section 11(a):

(a) Confidentiality Obligations. Confidential Information will not be disclosed or made available by the receiving party, directly or indirectly, to any third party, except as shall be agreed to in writing by the disclosing party. Each of the parties agrees to take all reasonable steps to preserve the confidentiality of the other's Confidential Information in accordance with their respective policies for the protection of their own non-public information (which policies shall provide at least reasonable protection) and agrees that it will be made available only to those employees and contractors as shall have a need to see and use for the purpose of fulfilling that party's obligations under this R&D Agreement, and any such employee and contractor shall be informed of the confidential nature of the Confidential Information and shall be required to observe the confidentiality obligations in respect thereof. The receiving party shall ensure that all Confidential Information received by it is kept separate (together with all information generated by the receiving party therefrom) from all documents and other records of the receiving party, and that it shall not use,

reproduce, transfer or store any of the Confidential Information in an extremely accessible place.

(8) Section 15 of the R&D Agreement is hereby amended by adding the following clause at the end: provided, however, that this R&D Agreement, in its entirety, shall be assignable by Opnext (or any successor to Opnext) to any Wholly-Owned Subsidiary of Opnext. For the avoidance of doubt, the parties agree that an Initial Public Offering (as defined in the Stockholders' Agreement) shall not require the consent of Hitachi.

(9) Section 28 of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following:

**Section 28. Survival.** To the extent the terms of this R&D Agreement provide for rights, interest, duties, claims, undertakings and obligations subsequent to the termination or expiration of this R&D Agreement, such terms of this R&D Agreement shall survive such termination or expiration, including but not limited to the terms of Sections 1, 4, 5 (with respect to licenses to Intellectual Property (other than Licensed Hitachi Other R&D IP) granted hereunder prior to the effective date of such termination), 6, 8, 9 (with respect to accrued but unpaid expenses), 11, 12, 18, 19, 20, 21, 22, 23, 26-34.

(10) Section 31 of the R&D Agreement is hereby amended by deleting it in its entirety and replacing it with the following:

**Section 31. Bankruptcy.** The parties agree that if a party becomes a debtor or debtor-in-possession under Title 11 of the United States Code (the "Bankruptcy Code"): (i) in the event of a rejection or proposed rejection of this R&D Agreement under Section 365 of the Bankruptcy Code, any and all rights licensed pursuant to this R&D Agreement shall be deemed to fall within the definition of "intellectual property" under Section 101 of the Bankruptcy Code and, in connection therewith, Section 365(n) of the Bankruptcy Code shall be implicated by such rejection or proposed rejection; and (ii) notwithstanding Section 365(c) of the Bankruptcy Code or applicable non-bankruptcy law which prohibits, restricts or conditions the assignment or assumption of this R&D Agreement or any of the rights therein, but subject to the debtor-in-possession or trustee, as applicable, otherwise complying with the requirements of Section 365 of the Bankruptcy Code for assumption, the debtor-in-possession or trustee in bankruptcy may assume this R&D Agreement. The parties agree that if a party files for bankruptcy under the laws of any other jurisdiction, the terms of this section will apply to the extent necessary to preserve the rights provided in this Section 31.

(11) A new Section 34 is hereby added to the R&D Agreement which provides as follows:

**Section 34. Injunctive Relief.** Each party acknowledges and agrees that the other party's Intellectual Property and Confidential Information are valuable property of such other party and that a material breach of this R&D Agreement (including unauthorized use of Intellectual Property or disclosure of Confidential Information) will cause irreparable injury for which the injured party does not have an adequate remedy at law and for which monetary remedies are not sufficient. Each party shall be entitled to seek equitable relief (including the granting of injunctive relief in that party's favor) without the obligation of posting a bond if the other party makes or threatens a material breach of this R&D Agreement (including unauthorized use of Intellectual Property or disclosure of Confidential Information). Each party agrees that equitable relief is not exclusive of other remedies to which the other party may be entitled at law or in equity as a result of any such material breach of this R&D Agreement (including any unauthorized use or disclosure of that party's Intellectual Property or Confidential Information).

**Section 3. No Other Amendments.**

Except as expressly set forth herein, all other terms and conditions of the R&D Agreement shall remain unmodified, in full force and effect and shall apply to this Amendment.

**Section 4. Governing Law.**

This Amendment shall be governed by and construed in accordance with the laws of Japan without giving effect to any choice-of-law or conflict-of-law provision or rule (whether of Japan or any other jurisdiction) that would cause the application of the laws of any jurisdiction other than of Japan. Regardless of the law applied, because this contract is in English, the terms and conditions of this contract will be interpreted in accordance with the meaning of the words in American colloquial English, notwithstanding any meaning of any word when translated into its Japanese equivalent.

**Section 5. Dispute Resolution.**

In the event of any dispute under this Amendment, as a condition precedent to either party seeking arbitration, in connection therewith, the parties will attempt to resolve such dispute by good faith negotiations (except for actions seeking injunctive relief). Such negotiations shall first involve the individuals designated by the parties as having general responsibility for the R&D Agreement. If such negotiations do not result within thirty (30) days from written notice of either party indicating that a dispute exists (a "Dispute Notice") in a resolution of the dispute, Opnext shall nominate one (1) corporate officer of the rank of vice president or higher and Hitachi shall nominate one (1) corporate officer of the rank of Board Director or higher, which corporate officers shall meet in person and attempt in good faith to negotiate a resolution to the dispute. In the event the corporate executives are unable to resolve the dispute within forty-five (45) days of receipt by either party of a Dispute Notice, a party may refer the matter to arbitration (except in the case of disputes arising under Section 11(c) or Section 34 of the R&D Agreement for which the parties may seek injunctive relief). In the event

that either party submits the dispute to arbitration, both parties shall cooperate in such binding arbitration in accordance with Exhibit C to the R&D Agreement.

**Section 6. Interpretation.**

The headings and captions contained in this Amendment and in any Exhibit are for reference purposes only and do not constitute a part of this Amendment. The use of the word “including” herein shall mean “including without limitation.”

**Section 7. Severability.**

Whenever possible, each provision of this Amendment shall be interpreted in such manner as to be effective and valid under applicable law, but if any provision of this Amendment held to be invalid, illegal or unenforceable in any respect under any applicable law or rule in any jurisdiction, such invalidity, illegality or unenforceability shall not affect the validity, legality or enforceability of any other provision of this Amendment in such jurisdiction or affect the validity, legality or enforceability of any provision in any other jurisdiction, but this Amendment shall be reformed, construed and enforced in such jurisdiction as if such invalid, illegal or unenforceable provision had never been contained herein.

**Section 8. Order of Precedence.** To the extent of a conflict between this Amendment and the First Amendment, the terms and conditions of this Amendment shall control.

**Section 9. Counterparts.**

This Amendment may be executed in one or more counterparts, each of which shall be an original and all of which taken together shall constitute one and the same agreement.

\* \* \* \* \*



IN WITNESS WHEREOF, the parties have caused this Amendment to be executed by their duly authorized officers as of the Amendment Date.

**OPNEXT, INC.**

**HITACHI, LTD.**

/s/ Harry L. Bosco  
Name: Harry L. Bosco  
Title: President and Chief Executive Officer

/s/ Naoya Takahashi  
Name: Naoya Takahashi  
Title: Vice President and Executive Officer

**SIGNATURE PAGE TO SECOND AMENDMENT TO OPNEXT, INC. R&D AGREEMENT**